HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Managed aircraft noise complaints and flight track deviations by analysing large datasets from a dedicated monitoring system, producing detailed technical reports for operational irregularities.
- Coordinated the deployment and maintenance of **permanent and temporary Noise Monitoring Terminals (NMTs)** across multiple sites, including liaison for site access and installation logistics.
- Prepared regulatory submissions and environmental reports, including responses to audits and inspections, ensuring compliance with aviation standards and data management requirements like **GDPR**.
- Supported community engagement initiatives by maintaining and updating public-facing noise information resources, including the **Noise Information Portal** and airport noise webpages.
- Developed **Python scripts** to automate data extraction and processing from operational logs, reducing manual effort for noise event analysis by approximately **30%**.
- Designed and implemented **REST APIs** using **Java Spring Boot** to integrate disparate data sources for the flight tracking system, improving data availability for downstream analysis.
- Maintained and enhanced **noise, flight tracking, and complaint databases**, supporting performance monitoring and continuous improvement initiatives using **SQL** and data modelling techniques.

SKILLS

Data Analysis & Reporting – Aircraft Noise Monitoring, Flight Track Analysis, Complaint Investigation, Environmental Reporting, Regulatory Compliance, Technical Reporting, Data Interpretation, Performance Monitoring

Tools & Systems – Power BI, Noise Monitoring Systems, GIS Software, Microsoft Office Suite (Excel, Word, PowerPoint), WebTrak, Noise Information Portal

Programming & Scripting – Python, SQL, Java, Spring Boot

Database Management – PostgreSQL, MySQL, Oracle, Data Warehousing, Data Modelling

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker

Methodologies & Practices – Agile, Scrum, CI/CD, Jenkins, Git, Jira, Confluence, Change Management, GDPR

PROJECTS

Aviation Noise Operations Dashboard Enhancement

- Developed interactive **Power BI dashboards** to visualise aircraft noise performance and flight track deviations, enabling stakeholders to quickly identify trends and operational issues.
- Refactored SQL queries and implemented strategic indexing in the flight tracking database, reducing analytical report execution times by approximately **22%**.
- Collaborated with operations teams to gather requirements for new reporting features, ensuring the dashboards met the needs for regulatory compliance and performance monitoring.

Automated Environmental Data Ingestion Pipeline

- Built a **Python ETL pipeline** to ingest and process data from various environmental monitoring systems, consolidating information for regulatory reporting and analysis.
- Integrated the pipeline with **AWS S3** for data storage and retrieval, handling approximately **1,500 data files** per week.
- Wrote unit tests using **pytest** to ensure data integrity and pipeline reliability, achieving over **90% code coverage** for critical processing modules.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)